

Phil Salin (1949–1991) was an economist, entrepreneur, and intellectual whose interdisciplinary work bridged the fields of Austrian economics and computer science, directly influencing the development of smart contracts and market-based computing 1, 2. He is credited with providing the conceptual foundations for "agoric" systems—decentralized networks that coordinate through market mechanisms—and for being the first to formally argue that software code is a form of constitutionally protected speech 3, 4.

Aerospace and Early Innovations

In the early 1980s, Salin founded and served as the first president of **Arc Technologies/StarStruck**, which was the first U.S. company to design, develop, and launch a privately financed rocket 1, 5. His efforts in this field played a significant role in the privatization of the satellite-launching industry 1.

Prior to his work in aerospace, Salin demonstrated early technical ingenuity while working as a programmer at Bechtel Financing Services in the mid-1970s, where he developed a precursor to the modern spreadsheet 1. He also worked as a stockbroker and as an economist at a think tank, where he contributed to the policy groundwork that led to the breakup of the AT&T telecommunications monopoly 1, 5.

AMIX and the Agorics Project

In 1984, Salin founded the **American Information Exchange Corporation (AMIX)**, which served as the practical embodiment of his theories regarding information markets 6. AMIX, which became an Autodesk subsidiary in 1988, allowed users to bid on documents, research, and consulting, and is recognized as the first smart contracting system for negotiation and asset exchange 6, 7.

Salin was a pivotal figure in the **Agorics Project**, an interdisciplinary initiative launched in September 1989 at George Mason University 8, 9. He is noted for introducing computer scientists, including Mark S. Miller and K. Eric Drexler, to the "spontaneous order" theories of F.A. Hayek 10, 11. This intellectual cross-pollination led to several foundational concepts:

- **Agoric Computing:** The application of price systems and market bidding to manage scarce computational resources like processor time and memory 12, 13.
- **Incentive Engineering:** Designing algorithms to ensure that the cost of using a computational object reflects its real value, encouraging efficient software evolution 13, 14.
- **Property Rights in Code:** The realization that object-oriented programming principles—specifically encapsulation—function as a digital version of property rights, allowing independent computational plans to operate without central coordination 15, 16.

Freedom of Speech in Software

In July 1991, Salin submitted a landmark position paper to the U.S. Patent and Trademark Office titled "Freedom of Speech in Software" 3. He argued that computer programs are a form of writing and should be protected by the First Amendment 3, 17. He contended that software patents were an "administrative fiction" that granted government monopolies over ideas, which

he equated to literary censorship 17, 18. This paper is cited as the first formal application of free speech concepts to software, a principle that would later become a core tenet of the Cypherpunk movement 4, 19.

Legacy and Influence on Cryptocurrency

Salin's legacy is defined by his vision of a "worldwide cryptographic commerce" system, a concept he articulated as early as 1988 20. He predicted that falling information and transaction costs would lead to a global information economy by the year 2000, where individuals could buy and sell information across borders as easily as they could domestically 21, 22.

His work influenced the birth of cryptocurrency and decentralized markets in several ways:

- **Conceptual Precursor:** His Agoric paradigm provided the theoretical scaffolding for decentralized systems that function through economic incentives rather than central authority 13, 23.
- **Influence on Cypherpunks:** Prominent Cypherpunks, such as John Gilmore, cited Salin's arguments on software patents and free speech as the first time these issues were linked, galvanizing the movement to protect code from government overreach 4, 19.
- **Smart Contracts:** His work on AMIX and the subsequent development of the Joule and E programming languages (which followed Agoric principles) laid the groundwork for the Object-Capability (OCAP) security model and modern smart contract frameworks like Zoe 7, 24, 25.